DO NOW, AO1 REVISION

Energy stores and systems

Energy
Paper 1, Combined and Separate
29 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 14 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	**Name the eight stores of energy.**	Chemical, electrostatic, magnetic, elastic potential, nuclear, thermal, kinetic and gravitational potential	☐
2 ★	**State the law of conservation of energy.**	Energy can be transferred usefully, stored or dissipated, but it cannot be created or destroyed.	☐
3 ★	**What is a closed system?**	A system where no energy enters or leaves.	☐
4 ★	**Define work done.**	The energy transferred when a force moves an object through a distance.	☐
5 ★	**State the standard unit of energy.**	Joule (J)	☐
6 ★	**Name the store of energy associated with fuels and batteries.**	The chemical energy store	☐
7	**Describe the change in the chemical energy store as fuel is burned.**	It decreases.	☐
8	**Name the store of energy associated with changes in speed.**	The kinetic energy store	☐
9	**Describe the change in the kinetic energy store as an object slows down.**	It decreases.	☐
10 ★	**Name the store of energy associated with changes in height.**	The gravitational potential energy store	☐
11	**Describe the change in the gravitational potential energy store as an object rises.**	It increases.	☐
12 ★	**Name the store of energy associated with extending and compressing objects.**	The elastic potential energy store	☐

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	QUESTION	ANSWER	✓
13	**Describe the change in the elastic potential energy store as a spring is compressed further.**	It increases.	☐
14 ★	**Name the store of energy associated with changes in temperature.**	The thermal energy store	☐
15	**Describe the change in the thermal energy store as an object gets hotter.**	It increases.	☐
16 ★	**Name the four ways energy is transferred between stores.**	Mechanically, electrically, by heating and by radiation	☐
17	**Name the energy transfer when a force moves an object.**	Mechanical transfer	☐
18	**Name the energy transfer when an object emits light or infrared.**	Transfer by radiation	☐
19	**Name the energy transfer when a hot object warms a cooler one.**	Transfer by heating	☐
20	**What is a system?**	An object or a group of objects.	☐
21	**State what happens to energy when a system changes.**	It is transferred from one store to another.	☐
22	**State the equation linking kinetic energy, mass and speed.**	kinetic energy = 0.5 × mass × speed^2 ($E_k = ½ m v^2$)	☐
23	**State the equation linking gravitational potential energy, mass, gravitational field strength and height.**	gravitational potential energy = mass × gravitational field strength × height ($E_p = m g h$)	☐
24	**State the equation linking elastic potential energy, spring constant and extension.**	elastic potential energy = 0.5 × spring constant × extension^2 ($E_e = ½ k e^2$)	☐
25 ★	**Name the two quantities that kinetic energy depends on.**	Mass and speed	☐
26	**Name the three quantities that gravitational potential energy depends on.**	Mass, gravitational field strength and height	☐

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	QUESTION	ANSWER	✓
27 ★	**A ball is thrown straight up. Describe the changes in its energy stores as it rises.**	Kinetic energy store decreases; gravitational potential energy store increases.	☐
28 ★	**An electric kettle is switched on. Describe the useful energy transfer.**	Energy is transferred electrically to the thermal energy store of the water.	☐
29 ★	**An apple falls from a tree. Describe how its energy stores change as it falls, before it hits the ground.**	Energy is transferred from the gravitational potential energy store of the apple–Earth system to the kinetic energy store of the apple.	☐

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